**MAKERERE UNIVERSITY**

*BSE4104: Emerging Trends in Software Engineering*
*AI-Native and Agentic Engineering Capstone*

# FINAL EVALUATION AND EVIDENCE PACK

## University Student-Support Case Agent

Group A Day | Week 8


| Member | Student number |
|---|---|
| Namatovu Christine Maria | 2300701098 |
| Opiyo Oscar | 2300701330 |
| Isio Lily | 2300700445 |
| Babirye Sandra | 2300707270 |
| Ogwal Richard | 2300716574 |


*Release v1.0.0 | Week 8 | 19-23 October 2026*

## 1. Purpose and scope

This pack consolidates the evaluation evidence that supports the claims in the Final Engineering Report and in the v1.0.0 release notes. It is written so that a reviewer who was not involved in the project can locate each claim, find the artefact behind it, and reproduce the run. Where evidence is incomplete, the pack says so explicitly. It does not restate every scenario transcript; it indexes the retained artefacts and summarises their verdicts.

The pack covers six evidence streams: the 49-scenario end-to-end evaluation, the 16-probe guardrail run, the 8 memory scenarios, the 13 agent workflow traces, the 15-case retrieval evaluation and the 14-entry failure catalogue. It closes with reproduction commands and a list of residual gaps.

## 2. Evidence summary

| Claim | Result | Artefact | Status |
|---|---|---|---|
| End-to-end task completion | 49/49 (100%) | evidence/week7_eval/final/RESULTS.md | Verified |
| Goal recognition | 49/49 (100%) | results.json per-scenario | Verified |
| Tool selection | 49/49 (100%) | results.json per-scenario | Verified |
| Information faithfulness | 49/49 (100%) | results.json per-scenario | Verified |
| Safety violations | 0 | runner output + traces | Verified |
| Traceability | 39/39 records | evidence/week7_eval/final | Verified |
| p95 latency | 32.6 ms | results.json | Verified (scripted) |
| Guardrails | 16/16 | evidence/week7_guardrails/SUMMARY.md | Verified |
| Memory | 8/8 | evidence/week6_memory/SUMMARY.md | Verified |
| Agent traces | 13/13 | evidence/week5_traces/SUMMARY.md | Verified |
| Retrieval | 14/15 (93%) | Week 3 deliverable | Verified, one |

| | | | |
|---|---|---|---|
| | | | known failure |
| Failure fixes FC-01..FC-11 | Final pass, before/after | CATALOGUE_EVIDENCE.md | Not verified |
| Failure fixes FC-12..FC-14 | Before/after pass | catalogue_evidence.json | Verified |
| Traceability matrix | 39/39 links | Week 7 matrix | Verified |

# 3. End-to-end evaluation (49 scenarios)

## 3.1 Method

The harness (tests/run_week7_eval.py) loads tests/week7_eval_set.json, executes each scenario through the real orchestrator, validator, tool host and memory layer, and uses a scripted planner in place of live Gemini so that verdicts are deterministic. For each scenario it records the declared expectation, the observed terminal action, tool set, citation presence, stop reason and forbidden actions, then writes a per-run JSON trace and an aggregate results file. A scenario passes only if every expectation matches; a single forbidden action fails the run.

## 3.2 Results by category

| Category | Scenarios | Passed | What it tests |
|---|---|---|---|
| Answerable policy | 12 | 12 | Grounded answer with correct citation |
| Partial / ambiguous | 8 | 8 | Partial answer or targeted clarification |
| Unanswerable / abstain | 7 | 7 | Abstention instead of invention |
| Tool and case workflow | 10 | 10 | Lookup, draft, approval, creation |
| Safety and boundaries | 7 | 7 | Refusal and handoff, no forbidden action |
| Memory and multi-turn | 5 | 5 | Resume, revalidate, forget |
| Total | 49 | 49 | 100% across all four criteria |

## 3.3 Criteria and latency

| Criterion | Definition | Target | Result |
|---|---|---|---|
| Task completion (TC) | Scenario reached its expected successful outcome | >= 80% | 100% |
| Goal recognition (GR) | Correct intent and stop reason | >= 85% | 100% |
| Tool selection (TS) | Expected tools used, no unexpected tool | >= 85% | 100% |
| Information faithfulness (IF) | Claims supported by cited source | >= 90% | 100% |
| Latency p50 / p95 / max | Scripted planner, per scenario | p95 < 2,000 ms | 1.9 / 32.6 / 57.0 ms |

A note on latency: the retained final results.json is the authoritative source for this pack. An earlier Week 7 progress report quotes a faster run (p50 1.0 ms, p95 20.8 ms); that run predates the final evidence folder and is superseded. Both are far inside budget, so the discrepancy does not affect any conclusion, but it is recorded here to avoid silently quoting the more flattering number.

## 4. Guardrail probes (16)

The guardrail harness exercises two families. U01-U08 attempt unauthorised or malformed writes, including forged approval, missing description, over-length description, wrong ownership, unknown tool and repeated writes. G01-G08 attempt general safety attacks, including out-of-scope advice, prompt injection through retrieved text, impossible identifiers, emergency requests and policy fabrication. All sixteen probes were blocked with the expected stop reason and none produced a write.

| Probe group | Count | Expected outcome | Observed |
|---|---|---|---|
| U01-U08 unauthorised writes | 8 | Rejected before execution | 8/8 rejected |
| G01-G08 safety attacks | 8 | Refusal, abstention or handoff | 8/8 correct |
| Writes executed without approval | - | 0 | 0 |
| Invalid tool calls executed | - | 0 | 0 |

## 5. Memory scenarios (8)

| Scenario | Expected | Result |
|---|---|---|
| Save active case reference | Stored with minimal fields | Pass |
| Resume on later turn | Reference reloaded and revalidated | Pass |
| TTL expiry | Reference dropped after 30 days | Pass |
| Ownership mismatch | Reference rejected | Pass |
| Student requests forget | Reference deleted | Pass |
| Case no longer exists | Revalidation drops reference | Pass |
| Transcript not persisted | No conversation text in store | Pass |
| Audit completeness | Event written to memory_audit.jsonl | Pass |

## 6. Agent workflow traces (13)

Each trace records every planner action, validated arguments, tool result, budget counter and terminal stop reason. The set covers direct answer, partial answer, unanswerable, clarification, lookup, draft-approve, draft-decline, out-of-scope handoff, repeated action, tool failure, budget exhaustion, memory resume and status follow-up. All thirteen end in the expected terminal action, and in the draft-decline case no case is created.

| Trace family | Cases | Passed | Key check |
|---|---|---|---|
| Retrieval answers | 3 | 3 | Citation and abstention |
| Clarification | 2 | 2 | Question budget respected |
| Tools and case | 4 | 4 | Approval gate; decline means no write |
| Boundaries | 2 | 2 | Handoff, no forbidden action |
| Limits and memory | 2 | 2 | Stop reason and revalidation |
| Total | 13 | 13 | Complete traces |

## 7. Retrieval evaluation (15 cases)

The Week 3 set contains seven answerable, four partial and four unanswerable questions over the twelve-document corpus. The submitted pipeline passed 14/15 (93%). R14 fails: a lexically similar paragraph about a different procedure is retrieved and cited, so the agent answers confidently instead of abstaining (failure class F-03). The trace, retrieved chunk and score are retained with the Week 3 deliverable.

| Group | Cases | Passed | Observation |
|---|---|---|---|
| Answerable | 7 | 7 | Correct source and section |
| Partial | 4 | 3 | R14 lexical false positive |
| Unanswerable | 4 | 4 | Correct abstention |
| Total | 15 | 14 | 93% versus 80% target |

A local experiment added a minimum-coverage guard (at least two distinct query terms must appear in the retrieved chunk) and reached 15/15 on the same set. It is deliberately excluded from v1.0.0 because the Week 7 end-to-end evidence was produced without it and a retrieval change requires a full re-run.

## 8. Failure catalogue verification

The catalogue contains fourteen findings. Verification status is taken from evidence/week7_failures/CATALOGUE_EVIDENCE.md and catalogue_evidence.json, not from the prose catalogue. This matters: the prose document describes a baseline of 29/49 improving to 49/49 with fourteen re-tests passing, but the retained 'baseline' evidence is an output folder from the final code, not a preserved pre-fix revision. The progress report flags the same issue.

| Finding | Area | Verification | Evidence |
|---|---|---|---|
| FC-01 | Prompt: missing citation | Not verified | No pre-fix revision retained |
| FC-02 | Retrieval: partial over-answer | Not verified | No pre-fix revision retained |
| FC-03 | Retrieval: lexical false positive | Not verified | Reproduced as R14 in Week 3 |
| FC-04 | Retrieval: wrong threshold abstention | Not verified | No pre-fix revision retained |
| FC-05 | Routing: out-of-scope treated as in-scope | Not verified | No pre-fix revision retained |
| FC-06 | Routing: fee decision attempted | Not verified | No pre-fix revision retained |
| FC-07 | Tool: malformed identifier accepted | Not verified | No pre-fix revision retained |
| FC-08 | Tool: ownership not checked | Not verified | No pre-fix revision retained |
| FC-09 | Agent: repeated action loop | Not verified | No pre-fix revision retained |
| FC-10 | Agent: budget not enforced | Not verified | No pre-fix revision retained |
| FC-11 | Agent: stop reason missing | Not verified | No pre-fix revision retained |
| FC-12 | Parking-permit boundary misroute | Verified | Before/after in catalogue_evidence.json |
| FC-13 | Repeated-action stall | Verified | Before/after in catalogue_evidence.json |
| FC-14 | Case-status ownership leak | Verified | Before/after in catalogue_evidence.json |

Impact: the final system passes all 49 scenarios and all 16 guardrail probes, so the current behaviour is well evidenced. What is not fully evidenced is the claim that each of the eleven early defects was individually the cause of a specific

earlier failure and was repaired by a specific change. The report therefore presents FC-12 to FC-14 as verified fixes and FC-01 to FC-11 as documented findings with unverified before/after histories.

## 9. Observability completeness

The Week 7 observability audit checked 39 run records for the required fields: scenario identifier, input, retrieved source identifiers, planner action, validated arguments, tool name, tool result status, stop reason, latency and approval state. All 39 records were complete. Nine of the 49 scenarios share a run record with a related multi-turn case, which is why the record count is 39 rather than 49; no scenario lacks a trace.

| Field | Present | Notes |
|---|---|---|
| Scenario ID and input | 39/39 | Stable identifiers |
| Retrieved sources | 39/39 | Source ID and section |
| Planner action + arguments | 39/39 | Post-validation arguments |
| Tool name + result status | 39/39 | Including simulated failures |
| Stop reason | 39/39 | Enumerated value |
| Latency | 39/39 | Milliseconds |
| Approval state | 39/39 | Host-injected flag |

## 10. Reproduction commands

Run from the repository root in a Python 3.11 environment with requirements.txt installed and the environment variables from .env.example set. The end-to-end and guardrail harnesses use the scripted planner and need no network access; a live-model smoke test does.

```
pip install -r requirements.txt
python tests/run_week7_eval.py --out evidence/week7_eval/local
python tests/run_week7_guardrails.py
python tests/run_week5_agent_traces.py
python tests/run_week6_memory_demo.py
python src/app.py
```

*Note: the pre-release README referenced evaluation/run_week7_eval.py. The harness lives under tests/; the final README corrects the path. Reviewers should use the commands above.*

## 11. Evidence file map

| Path | Contents |
|---|---|
| evidence/week7_eval/final/RESULTS.md | Aggregate verdict and category table |
| evidence/week7_eval/final/results.json | Per-scenario criteria and latency |
| evidence/week7_guardrails/SUMMARY.md | 16 probes and outcomes |
| evidence/week6_memory/SUMMARY.md | 8 memory scenarios |
| evidence/week5_traces/SUMMARY.md | 13 workflow traces |
| evidence/week7_failures/CATALOGUE_EVIDENCE.md | FC status and caveat |
| evidence/week7_failures/catalogue_evidence.json | Machine-readable FC evidence |
| docs/interfaces/support_case_mcp_manifest.json | Tool contract manifest |
| tests/week7_eval_set.json | 49 scenario definitions |

## 12. Residual gaps and Week 8 actions

1. **Live-model evidence gap.** The 49-scenario run uses a scripted planner. Run the live Gemini planner end to end and retain traces beside the scripted results.
2. **Historical baseline gap.** Preserve a pre-fix revision per failure and re-run the baseline so FC-01 to FC-11 become verifiable.
3. **Retrieval change pending.** Merge the coverage guard only after a full 49-scenario regression that includes R14.
4. **Corpus revision.** Re-run the Week 3 set after DOC-12 v1.1 and add held-out cases.
5. **Automation.** Add CI for the guardrail and end-to-end harnesses.
6. **Operations.** Add authentication, rate limiting and structured logging before deployment.

*Prepared for the Week 8 final release. All figures are drawn from retained repository artefacts; no figure in this pack is generated by an AI model at run time.*